MIL-P-14321E
30 May 1984
SUPERSEDING
MIL-P-14321D
15 June 1976

MILITARY SPECIFICATION

PUMP, CENTRIFUGAL: FOR PETROLEUM PRODUCTS, DIESEL-ENGINE-DRIVEN, SELF PRIMING, SKID-MOUNTED, SINGLE STAGE; 6-INCH, GROOVED ADAPTERS; 245 GPM AT 205 FEET HEAD TO 1120 GPM AT 100 FEET HEAD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a skid-mounted, diesel-engine-driven, direct-connected (through a flexible coupling), self-priming centrifugal pump, with the necessary accessorie' and connections, for petroleum products.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Command, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4320

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

QQ-C-390	- Copper Alloy Castings (Including Cast Bar).
UU-T-81	- Tags, Shipping and Stock.
WW-F-406	- Flanges, Cast-Iron (Classes 125 and 250) and Bronze (Classes 150 and 300).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-P-40	- Packaging and Packing of Hand Tools.

MILITARY

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-R-196	- Repair Parts, Accessories, and Kits, Mechanical: Packaging of.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2638	- Lacquer, Vinyl Resin, Gasoline and Water Resistant.
MIL-D-2940	- Dampener, Fluid Pressure, Gage Protection.
MIL-C-3774	- Crate, Wood; Open, 12,000- and 16,000-Pound Capacity.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-P-10388	- Pipe Fittings: One or More Ends Grooved.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-T-22085	- Tapes, Adhesive, Preservation and Sealing.
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-I-24137	- Iron Castings, Nodular Graphitic (Ductile Iron) and Nodular Graphitic (Corrosion Resisting, Austenitic, Low Magnetic Permeability) (for Shipboard Application).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-S-52506	- Seal, Shaft, Mechanical, Low Pressure Type, Self Lubricating, for General-Purpose Centrifugal Pumps.
MIL-M-52590	- Muffler, Exhaust, Internal Combustion Engine, General Purpose.
MIL-E-52649	- Engine Cold Starting Aids, Ether Fuel Primer.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS35643	- Gasket: Synthetic Rubber, Cap, Fuel Tank.
MS35644	- Filler Neck: Fuel Tank - Military Vehicles.
MS35645	- Cap, Fuel Tank: Tactical Vehicle, Fording.
MS35829	- Hinge, Butt, Continuous (Piano), .075 Thickness, .187 and .250 Pin Dia.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

MS39247	- Air Cleaner, Intake; Dry Type, for Internal Combustion Engines (Class I and II).
MS53063	- Indicator, Air Cleaner, Intake; Restriction Mechanical Type.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

5100-1A United States Department of Agriculture Forest Service Standard.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME	- Boiler and Pressure Vessel Code.
ANSI/AWS D 1.1	- Structural Welding Code - Steel.
ANSI B 40.1	- Gauges-Pressure and Vacuum, Indicating Dial Type Elastic Element.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 36	- Structural Steel.
ASTM A 48	- Gray Iron Castings.
ASTM A 276	- Stainless and Heat-Resisting Steel Bars and Shapes.
ASTM A 331	- Steel Bars, Alloy, Cold-Finished.
ASTM D 2000	- Rubber Products in Automotive Applications.
ASTM D 3951	- Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 East Flair Drive, El Monte, CA 91731.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

HYDRAULIC INSTITUTE

Rating Standards and Test Code Standards.

(Application for copies should be addressed to the Hydraulic Institute, 122 East 42nd Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The pump shall consist of a skid-mounted, diesel-engine-driven, self-priming pump with grooved adapters (hereinafter referred to as "pump unit"). The pump unit shall be as shown on Figure 1 and as specified herein.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for inspection and approval (see 4.4 and 6.3).

3.3 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.8).

3.3.1 Material deterioration and control. The pump unit shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Threaded parts. All screw threads shall be in accordance with FED-STD-H28.

3.5 Environmental.

3.5.1 Operating temperature. The pump unit shall perform as specified herein in any ambient temperature from plus 125° F to minus 25° F.

3.5.2 Storage temperature. The pump unit shall not be damaged by storage in any ambient temperature from plus 160° F to minus 50° F.

3.6 Safety. All shafts or rotating or moving parts of the pumping unit shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of the equipment, including exhaust pipes exposed to contact by personnel or which create a fire hazard, shall be fully guarded or insulated. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto the engine, exhaust, or electrical equipment. Exhaust or discharges from the equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard.

3.7 Human factors. The pump unit shall conform to the human factors engineering criteria as specified in MIL-STD-1472.

3.7.1 Noise limits. The noise produced by the Pump shall not exceed MIL-STD-1474 paragraph 5.1.1.1 Category D (85dB(A)), at the operators' position or other occasionally occupied positions, when tested in accordance with 4.6.2.6.1. If technical infeasibility is established via MIL-STD-1474 paragraph 5.1.1.2, or the means of reduction is too costly, the written permission of the procuring activity shall be requested to exceed the 85dB(A) limit. The precautions of MIL-STD-1474 (see 6.7) shall be provided if and only if MIL-STD-1474 paragraph 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity. Hazard signs shall conform with MIL-STD-1474 paragraph 4.3 and be readable at the distance indicated on the sign. Hearing protectors shall not be supplied with the pump units.

3.8 Reliability. The specified mean-time-between-failure (MTBF) shall be 2000 hours when the pump unit is operating at rated capacity and tested as specified in 4.6.2.7.

Source: https://assist.dla.mil -- Downloaded: 202610-18T03:03Z
Check the source to verify that this is the currentversion before use.

3.9 Maintainability.

3.9.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, and lubrication charts, as specified (see 6.5 and 6.6), shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.6.2.8.

3.9.2 Maintenance ratio. The pump unit shall have a maintenance ratio of not more than 0.07. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.10 Performance. The pump unit shall prime and deliver not less than 1120 gallons per minute (gpm) within 2 minutes when operated with a static suction lift equivalent to 10 feet of water at an atmospheric pressure of 29.92 inches of mercury. The pump unit shall develop a peak efficiency of not less than 74 percent at the rated condition points. The maximum net positive suction head (NPSH) requirement shall not exceed 12 feet at 1120 gpm at 100 feet total head at the operating speed of the pump unit. When operating at a maximum continuous speed of 2600 revolutions per minute (rpm) with a static suction lift equivalent to 10 feet of water at a barometric pressure of 29.92 inches of mercury, the pump unit shall deliver not less than the following:

a. 245 gpm at 205 feet total head.

b. 729 gpm at 180 feet total head.

c. 1120 gpm at 100 feet total head.

All data shall be corrected to sea level atmospheric conditions of 29.92 inches of mercury and fuel having a specific gravity of 0.725.

3.10.1 Performance identification plate. A performance identification plate not less than 8 inches by 10-1/2 inches shall be permanently attached to the inside cover of the instrument panel. The chart shall show the complete performance characteristics of the pump unit when operating at the maximum continuous operating speed and horsepower and when pumping liquid fuel with a specific gravity of 0.725. All data shall be derived from the actual performance of the first article pump unit. The plate shall conform to MIL-P-514, type III, composition C of type I, grade A, class 1 material.

3.11 Pump. The pump shall be the vertically split case centrifugal type and shall be self priming after the case has been initially filled. The suction and discharge ports shall be 6 inches nominal, with plain face flanges conforming to WW-F-406, class 250. The pump shall be furnished with a mechanical shaft seal and shall be close coupled. An intermediate-type bearing housing fabricated of cast iron conforming to ASTM A 48 shall be directly mounted to the engine flywheel and the pump case. The intermediate housing shall be provided with antifriction,

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

lubricated-for-life or prepacked, sealed bearings of such size and type so as to absorb the maximum radial and thrust loading imposed by the pump impeller without permanent deformation or breakage. The pump shall be provided with a 6-inch flange-to-groove suction nipple conforming to MIL-P-10388, class 250, plain face and a suction check valve. The check valve gasket shall be fabricated from synthetic rubber conforming to ASTM D 2000, type 2BG515B14E61F17. The discharge shall be provided with a 6-inch flange-to-groove elbow. Flange and groove dimensions shall conform to MIL-P-10388, class 250. The nipple and elbow may be of either threaded or welded construction. The discharge elbows shall have a coupling welded on at the highest possible point for connection of a vent valve. The pump shall be capable of priming and pumping light-weight petroleum products and gasoline.

3.11.1 Hydrostatic pressure. The completely assembled pump, with seals, elbow, and vent valve in place, shall withstand an internal hydrostatic pressure of not less than 240 pounds per square inch (psi) for a period of not less than 2 minutes without any evidence of leakage through the pump or accessories.

3.11.2 Pump case. The pump case shall be iron castings conforming to ASTM A 48 or higher, stress-relief annealed. The pump case shall be provided with a lifting attachment above the center of gravity of the pump. The attachment shall withstand a load of not less than 2-1/2 times the weight of the pump. The case shall be provided with a 3/4-inch NPT drain port and plug at the low point of the case, and a provision for filling the pump case without removal of the suction or discharge connections.

3.11.3 Pump impeller and case wear rings. The pump impeller shall be material conforming to QQ-C-390, type I, copper alloy No. 946 or MIL-I-24137, class A. The case wear rings shall be material conforming to QQ-C-390, type I, copper alloy No. 946. The impeller shall be balanced while rotating at the continuous operating speed of the pump unit to an out-of-balance condition of not more than 0.25 ounce-inch (oz-in).

3.11.4 Impeller shaft. The impeller shaft shall be corrosion-resisting steel conforming to ASTM A 276, type 410, condition H, or steel conforming to ASTM A 331, grade designation 4140, cold finished. The shaft shall have a tensile strength of not less than 130,000 psi. All shoulders and corners shall be chamfered or beveled to avoid stress concentrations and damage to O-rings during seal installations.

3.11.5 Pump shaft seal. The pump shaft seal shall conform to MIL-S-52506, type IV. Provision shall be made for applying oil to all parts of the carbon insert so that the carbon is sealed from the atmosphere during shipment, storage, and idle periods.

3.11.6 Vent. A 1/2-inch globe valve with fittings and copper tubing shall be provided for manually venting the high point of the discharge elbow. The copper tubing shall extend from the globe valve to a point on top of the skid base that will direct the resultant discharge to the outside of the skid base. The valve and fittings shall withstand not less than 240 psi hydrostatic pressure without leakage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

3.11.7 Gages. Two vibration-resistant pressure gages of nonoxidizing material shall be provided. The discharge gage shall conform to ANSI B 40.1 and shall have a range from 0 to 300 psi pressure. The compound suction gage shall conform to ANSI B 40.1 and shall have a range from 0 to 100 psi pressure and a range from 0 to 30 inches mercury vacuum. Each gage shall be weatherproof and accurate to within 2 percent of the total gage reading. The dials shall be white with black calibration lines and figures, and shall be not less than 4-1/2 inches in diameter. The gages shall be located on the instrument panel and shall be provided with a fluid-pressure dampener conforming to MIL-D-2940, type II, class 1, composition optional. Shutoff cocks shall be provided in gage lines. The gage lines shall be provided with provisions for draining at the lowest point. The gage lines, including shutoff cocks, shall withstand full hydrostatic pressure of the pump without leakage.

3.12 Engine. For the purpose of selecting an engine, the pump unit shall be classified as a class II end item in accordance with MIL-STD-1410. The engine, including all systems, components, and accessories, shall be in accordance with MIL-STD-1410, except as otherwise specified herein, and shall be furnished with the following accessories:

a. Pusher-type cooling fan.

b. Spark-arresting muffler.

c. Manually operated engine fuel primer pump.

d. Engine cold-starting aid in accordance with MIL-E-52649, type III, electric.

e. Industrial-type footed flywheel housing.

f. Engine housing.

g. Engine stop control.

h. Instrument panel with hinged door and complete engine instrumentation (see 3.12.9), except that fuel level indicator is not required.

3.12.1 Engine operational requirements. The engine operational requirements and maximum engine operating limits shall be in accordance with MIL-STD-1410 under any of the requirements and conditions specified herein. The engine, as applied to the pump unit, shall meet the applicable exhaust emissions standards.

3.12.2 Engine drains. The engine crankcase, engine block water jacket, and radiator shall be provided with drain cocks and hoses to allow draining of the units outside the skid base. Drain hoses shall be retained in a protected location when not in use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

3.12.3 Fuel system. The fuel tank shall be fitted with a filler cap gasket, filler neck, and filler cap conforming to MS35643, MS35644, and MS35645, respectively. The interior of the fuel tank shall be treated in accordance with MIL-T-704 and then given a three-coat paint system conforming to MIL-L-2638. The tank shall be equipped with a dial-type, mechanical, liquid level gage with a 1-1/2-inch external pipe thread mounting and an automatic float-operated tank fill valve. The valve shall be 3/4-inch pipe size and shall operate on fuel line pressures from 1 to 25 psi. The tank shall have sufficient capacity for 4 hours of operation of the pump unit when the pump is operating at 1120 gpm at 100 feet total head. The tank shall be provided with a bottom shield fabricated of 3/16-inch steel plate conforming to ASTM A 36 for skidding protection. The tank outlet or fuel line shall be provided with a three-way selector valve and a 1/2-inch SAE flared fitting for connection with an auxiliary fuel supply. The tank and all lines shall be located or positioned to eliminate fire hazard caused by spilling, overflow, or drainage of fuel. Fuel filters shall be furnished and shall be mounted between the fuel tank and fuel pump.

3.12.4 Air induction system. The engine shall be equipped with a dry-type air cleaner conforming to MS39247, class I or II. The air cleaner shall be equipped with an air cleaner restriction indicator conforming to MS53063. The restriction indicator shall be mounted on the instrument panel and shall indicate when air cleaner service is required. The restriction indicator rating (MS dash number) shall be equal to the maximum allowable restriction (vacuum) recommended by the engine manufacturer, or equal to the nearest maximum signaling point (MS dash number) listed in MS53063.

3.12.5 Muffler. The spark-arresting muffler shall consist of a muffler conforming to MIL-M-52590 and a spark arrester conforming to USDA 5100-1A.

3.12.6 Battery-disconnect switch. A battery-disconnect switch shall be provided on the instrument panel to permit disconnection of the battery from the electrical system. The switch shall be in-line between the batteries and the cranking motor and shall withstand full cranking motor load.

3.12.7 Electric-cranking system. A 12-volt electric cranking system shall be furnished in accordance with MIL-STD-1410. The cranking system components shall be protected by an automatic starter disconnect system. The system shall automatically de-energize the starter or starter solenoid when the engine starts, and shall prevent starter engagement if the starter switch is activated while the engine is running. The cranking motor shall be furnished by the engine manufacturer or shall be approved by the engine manufacturer for this application.

3.12.8 Battery charging system. A 12-volt alternator battery charging system of not less than 35 ampere capacity shall be furnished in accordance with MIL-STD-1410.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

3.12.9 Wiring and instrumentation. All wiring, instrumentation, terminals, and connections shall be waterproofed. The instrument panel shall include the following instruments and controls:

a. Engine lube-oil pressure gage.

b. Coolant temperature gage.

c. Battery charging ammeter.

d. Tachometer (tachometer and drive shall be in accordance with SAE J678D, heavy duty).

e. Engine hourmeter.

f. Emergency manual engine control push-pull "kill" switch.

g. Air restriction indicator.

h. Engine starter switch.

i. Engine cold-starting aid control.

j. Manual engine-throttle control (vernier-type).

(Note: The engine hourmeter and tachmeter may be a combined instrument if a mechanically driven tachometer is furnished.)

3.12.10 Engine safety controls. An engine low oil pressure control and a high-temperature control shall be furnished and shall be the "shutoff" type in accordance with MIL-STD-1410.

3.13 Coupling. The pump shall be connected to the engine with a flywheel-type, dry-type, flexible coupling. The coupling shall transmit full engine torque and shall dampen torsional vibrations. The coupling shall be removable or replaceable without disconnecting the pump or engine from the skid frame or disconnecting the inlet or outlet piping.

3.14 Skid base. The pump unit shall be mounted on a pipeline-type skid base as shown in figure 1, and fabricated from structural steel conforming to ASTM A 36 with rounded ends for skidding or towing. The skid base shall be adequate to maintain alinement of the intermediate bearing housing, pump, diesel engine, and all accessories when tested as specified herein. The skid with pump, engine, and all accessories shall have a maximum ground pressure of 5 psi. No component of the pump unit, with the exception of the pump suction and discharge flange-to-groove adapters, shall overhang the skid base.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

3.15 Electromagnetic interference. The electromagnetic interference emission of the pump unit shall conform to MIL-STD-461, C1.

3.16 Lubrication.

3.16.1 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases as follows:

a. MIL-G-23827 - For ball, roller, and needle bearings, gears, and on sliding and rolling surfaces of such equipment as instruments, cameras, electronic gear, and mechanical control systems; for use on gears, actuator screws, and other equipment requiring a lubricant with high load-carrying capacity over a temperature range of minus 100° F to plus 250° F.

b. MIL-G-10924 - For lubrication and surface corrosion protection of automotive and artillery equipment operated over the temperature range of minus 65° F to plus 225° F. Also used in other applications within this temperature range where grease with oxidation-resistant and corrosion-preventive properties is desirable. Not for use on machinery which comes in contact with foods or potable water.

The pump unit shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not always compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.17 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.17.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

3.17.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442 South Coast Air Quality Management District.

3.18 Toolbox. The pump unit shall be provided with a weatherproof toolbox made of 0.075-inch nominal thickness steel. The toolbox shall hold the publications, operating and maintenance tools (see 6.5 and 6.6). The toolbox shall have an overlapping hinged lid to prevent the entry of rain or dirt. The lid shall be mounted on hinges conforming to MS35829 with a 0.187-inch-diameter brass pin. When a toolbox is mounted on the pump unit, the toolbox lid shall open to a minimum of 90 degrees from the horizontal and remain in the open position. The toolbox shall be provided with a latch that will secure the cover in the closed position. In addition, the toolbox shall be provided with a hasp for locking the box. The latch and hasp may be either one combined unit or two separate units. Two drainholes in opposite corners of the bottom of the box shall be provided. The toolbox shall be fastened to the pump unit skid base in a protected location.

3.19 Lifting attachment. The pump unit shall have a removable lifting bail located at the center of gravity of the pump unit to enable lifting from above. The lifting bail shall have an eye of not less than 3 inches inside diameter and shall be the highest point on the unit. The lifting bail shall withstand, without damage to any part of the pump unit, 2-1/2 times the resulting force imposed when the pump unit is suspended by the lifting bail.

3.20 Tiedown provisions. Permanently affixed tiedown provisions that permit fastening the pump unit to the floor or deck of a transportation medium shall be provided and shall conform to MIL-STD-209.

3.21 Identification marking. The pump units shall be identified in accordance with MIL-STD-130. The marking shall be applied to the pump unit on plates conforming to MIL-P-514, type I, style 1, composition C of type I, grade A, class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.22 Instruction plates. Each pump unit shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the pump unit. Instruction plates shall conform to MIL-P-514, type III, composition C of type I, grade A class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

3.23 Treatment and painting. The portions of the pump unit and its components and parts normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, color as specified (see 6.2).

3.24 Stenciling. The gross weight of the pump unit shall be stenciled on each side of the unit in Gothic capitals and the number shall be 2 inches high. Unless otherwise specified (see 6.2), stenciling shall conform to MIL-T-704.

3.25 Workmanship.

3.25.1 Castings and forgings. All parts, components, and assemblies of the pump unit which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Castings shall be sound and free from patching, misplaced coring, warping, porosity, or other defects which might render the casting unsound for its intended use.

3.25.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free from burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.25.3 Welders and welding.

3.25.3.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/ASME, Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D1.1, Structural Welding Code-Steel.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with ANSI/ASME or ANSI/AWS codes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

3.25.3.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the pump unit to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the pump unit shall be at the option of the contractor.

3.25.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than two full threads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Individual component inspection (see 4.3).

b. First article inspection (see 4.4).

c. Quality conformance inspection (see 4.5).

d. Inspection of packaging (see 4.7).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

4.3 Individual component inspection.

4.3.1 Engine. Prior to assembling the engine to the pump unit, each engine shall be subjected to the engine test marked "X" in column 2 of table I. Failure of the test shall be cause for rejection.

4.3.2 Pump assembly. Prior to assembling the pump to the engine, each pump complete with seals, elbow, and vent valve in place, shall be subjected to the hydrostatic test marked "X" in column 2 of table I. Failure of the test shall be cause for rejection.

4.3.3 Pump impeller. Prior to final assembling of the impeller in the pump, each impeller shall be subjected to the impeller test marked "X" in column 2 of table I. Failure of the test shall be cause for rejection.

4.4 First article inspection.

4.4.1 Examination. The first article pump unit shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. The first article pump unit shall be subjected to the tests marked "X" in column 1 of table I. Failure of any test shall be cause for rejection.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Individual. Each pump unit shall be examined for the critical defect specified in 4.6.1. Presence of a critical defect shall be cause for rejection.

4.5.2.2 Samples. Samples selected in accordance with 4.5.1 shall be examined for the major and minor defects specified in 4.6.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5.3 Tests.

4.5.3.1 Individual. Each pump unit shall be subjected to the tests marked "X" in column 2 of table I. Failure of any test shall be cause for rejection.

4.5.3.2 Samples. Samples selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in column 3 of table I. AQL shall be 2.5 percent defective.

Source: https://assist.dla.mil -- Downloaded: 20261O-18T03:03Z
Check the source to verify that this is the currentversion before use.

4.6 Inspection procedure.

4.6.1 Examination. The pump unit shall be examined as specified herein for the following defects:

Critical

1. Castings not as specified.

Major

101. Materials not as specified.
102. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Dimensions not as specified.
106. Parts or components missing or not as specified.
107. Screw threads not as specified.
108. Performance identification plate missing or not as specified.
109. Pump case lifting attachment missing or not as specified.
110. Impeller shaft not beveled or chamfered as specified.
111. Engine accessories missing or not as specified.
112. Engine drains not as specified.
113. Fuel system not as specified.
114. Air induction system not as specified.
115. Electric cranking system not as specified.
116. Wiring and instrumentation not as specified.
117. Coupling not as specified.
118. Part or parts of pump unit, except adapters, overhanging skid base.
119. Lubrication information tag missing.
120. Fungus and moisture resistance requirements not as specified.
121. Toolbox missing or not as specified.
122. Instruction plates missing, incorrect, or illegible.
123. Workmanship not as specified.

Minor

201. Identification marking missing, incorrect, or illegible.
202. Treatment and painting not as specified.
203. Stenciling missing or not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

4.6.2 Tests.

4.6.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed in accordance with the applicable provisions of the Test Code for centrifugal pumps of the Standards of the Hydraulic Institute. All performance data shall be corrected to standard sea level conditions of barometric pressure of 29.92 inches of mercury, and liquid fuel at 60° F having a specific gravity of 0.725. Water at a temperature of between 60° F and 80° F may be used as the test fluid.

4.6.2.2 Test schedule.

TABLE I. Test schedule.

First article	Quality conformance		Tests	Test paragraph	Requirement paragraph
	Individual	Sample			
1	2	3	4	5	6
X			High temperature	4.6.2.3	3.5
X			Low temperature	4.6.2.4	3.5
X			Safety	4.6.2.5	3.6
X			Human factors	4.6.2.6	3.7
X			Noise level measurement	4.6.2.6.1	3.7.1
X			Reliability	4.6.2.7	3.8
X			Maintenance support	4.6.2.8.1	3.9.1
X			Maintenance ratio	4.6.2.8.2	3.9.2
			Performance:		
X			Pump	4.6.2.9.1	3.10
X		X	Pump unit	4.6.2.9.2	3.10
X		X	Priming	4.6.2.9.3	3.11
X	X		Hydrostatic pressure	4.6.2.10	3.11.1
X	X		Impeller	4.6.2.11	3.11.3
X	X		Engine	4.6.2.12	3.12
X			Electromagnetic interference	4.6.2.13	3.15
X			Lifting attachment	4.6.2.14	3.19
X			Tiedown provisions	4.6.2.15	3.20

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

4.6.2.3 High temperature. The pump unit shall be tested as specified in MIL-STD-810, procedure I, method 501. The maximum ambient storage temperature shall be 160° F, and the maximum ambient operating temperature shall be 125° F. The operating period of Procedure I shall be 2 hours. The pump unit shall be operated at rated capacity. Nonconformance to 3.5 shall constitute failure of this test.

4.6.2.4 Low temperature. The pump unit shall be tested as specified in MIL-STD-810, procedure I, method 501. The minimum ambient storage temperature shall be minus 50° F. The minimum ambient operating temperature shall be minus 25° F. The operating period of procedure I shall be 2 hours. The pump unit shall be operated at rated capacity. Nonconformance to 3.5 shall constitute failure of this test.

4.6.2.5 Safety. The pump unit shall be evaluated throughout testing as specified in 4.6.2. Nonconformance to 3.6 shall constitute failure of this test.

4.6.2.6 Human factors. The human factors shall be evaluated during testing. Nonconformance to 3.7 shall constitute failure of this test.

4.6.2.6.1 Noise level measurement. Noise levels shall be measured in accordance with MIL-STD-1474, for steady-state noise. As a minimum: Noise levels shall be measured when equipment is operating at normal maximum rated load; measurements will be taken at the noisiest location, operators location, and at a 25-foot radius from the geometric center of the equipment every 45 degree ±5 degree of azimuth, 65±1 inches above the ground. Should the noise level exceed 85dB(A) at any noise test point, failure to comply with paragraph 3.7.1 or MIL-STD-1474 provisions shall constitute failure of this test.

4.6.2.7 Reliability. Using the MTBF specified in 3.8, the pump unit shall be tested as specified in 4.6.2.9 with "accept" and "reject" criteria in accordance with figure 2. Continue testing until either an "accept" or "reject" decision is reached. A failure is defined as any malfunction which cannot be corrected within 30 minutes (all waiting periods are excluded) by adjustment, repair, or replacement action using controls and on-equipment tools or parts available to the using element, and is a malfunction which causes or may cause one or more of the following:

a. Failure to commence operation, cessation of operation, or degradation of performance below specified levels.

b. Damage to the pump unit by continued operation.

c. Safety hazards to personnel.

Nonconformance to 3.8 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

4.6.2.8 Maintainability.

4.6.2.8.1 Maintenance support evaluation. Maintenance operations required during first article pump unit testing shall be accomplished to determine conformance to 3.9.1. Failure of the support elements to maintain the pump unit as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the MTBF.

4.6.2.8.2 Maintenance ratio. The maintenance ratio shall be computed during first article pump unit testing. Nonconformance to 3.9.2 shall constitute failure of this test.

4.6.2.9 Performance.

4.6.2.9.1 Pump. Prior to assembling the pump to the engine, operate the pump using a dynamometer or calibrated electric motor at the designed operating speed and at a static suction lift equivalent to 10 feet of water at standard conditions. Test the pump for the requirements specified in 3.10 in accordance with the Rating Standards and Test Code Standards for centrifugal pumps of the Hydraulic Institute. Data relative to head capacity shall be taken at a minimum of six points and shall include the rating points (see 3.10), point of maximum horsepower wide-open discharge, and shutoff. Delivery of less than the capacities specified in 3.10 with efficiency of less than 74 percent, an NPSH greater than 12, or a horsepower requirement greater than the continuous duty rating of the engine shall constitute failure of this test.

4.6.2.9.2 Pump unit (operational). Assemble the pump to the engine and operate the pump at the designed operating speed and at a static suction lift equivalent to 10 feet of water as standard conditions. Test the pump for the requirements specified in 3.10 in accordance with the Rating Standards and Test Code Standards for centrifugal pumps of the Hydraulic Institute. Data relative to head and capacity of the pump, and speed (rpm) of the engine, shall be measured and recorded at a minimum of six points on the pump performance curve. The measured points shall include the rating points (see 3.10), point of maximum horsepower, wide-open discharge, and shutoff. The pump unit shall run for a minimum of 5 minutes at each measured point. Delivery of less than the capacities specified in 3.10 or failure of the engine to meet any of the operational requirements or conditions specified in 3.12.1 shall constitute failure of this test.

4.6.2.9.3 Priming. Fill the pump case with either liquid petroleum fuel or a suitable solvent having a specific gravity of between 0.72 and 0.75, and conduct the priming test with a static suction lift equivalent to 10 feet of water at standard atmosphere conditions. This test may be performed lifting either water, gasoline of specific gravity between 0.725 and 0.75, or a solvent of specific gravity between 0.72 and 0.75. Inability of the pump to prime and deliver full capacity in 2 minutes shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202610-18T03:03Z
Check the source to verify that this is the current version before use.

4.6.2.10 Hydrostatic pressure. Apply an internal hydrostatic pressure of not less than 240 psi for a period of not less than 2 minutes to the completely assembled pump with seals, elbow, and vent valve in place. Nonconformance to 3.11.1 shall constitute failure of this test.

4.6.2.11 Pump impeller. Determine the out-of-balance condition of the pump impeller while rotating at the continuous operating speed of the pump unit. An out-of-balance condition of more than 0.25 oz-in shall constitute failure of this test.

4.6.2.12 Engine. Prior to assembling the engine to the pump unit, the engine shall be tested in accordance with the production engine tests and production control tests of MIL-STD-1410. Nonconformance to the requirements specified herein shall constitute failure of this test.

4.6.2.13 Electromagnetic interference. The first article pump unit shall be tested to determine conformance to 3.15. The contractor shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.6.2.14 Lifting attachment. Place the pump unit on a flat level surface and lift it with the lifting bail. Raise the pump unit until the skid has completely cleared the ground. With the pump unit in this position, measure the difference in elevation between the high and low ends of the skid. A difference in elevation greater than 12 inches shall constitute failure of this test.

4.6.2.15 Tiedown provisions.

4.6.2.15.1 Test procedure. With the pump unit anchored by means other than the tiedown being tested, apply the static loads specified in MIL-STD-209 to each provision. Measure and record the dimensions of the tiedowns.

4.6.2.15.2 Failure criteria. Weld failure, permanent deformation of the pump unit or its tiedown provisions, breakage of the provisions, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.7 Inspection of packaging.

4.7.1 First article pack inspection.

4.7.1.1 Examination. The first article pack shall be examined for the defects specified in 4.7.2.3. Presence of one or more defects shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

4.7.1.2 Test. The first article pack of the crated pump unit for level A or B shall be subjected to the rail impact test as specified in MIL-STD-1186, appendix A. At the conclusion of the first article pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking, or bracing, or any visual damage to the pump unit or components shall constitute failure of the first article pack and shall be cause for rejection.

4.7.2 Quality conformance inspection of pack.

4.7.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

124.	Materials, methods, or containers not as specified for level A or B. Each incorrect material, method, or container shall be considered one defect.
125.	Disassembly not as specified for level A.
126.	Matchmarking not as specified for level A.
127.	Matchmarking information not applied with waterproof ink as specified for level A.
128.	Unprotected surfaces not coated with preservative as specified for level A.
129.	Engine, components, or accessories not preserved as specified for level A.
130.	Pump units not preserved as specified for level A.
131.	Pump openings not sealed as specified for level A.
132.	Maintenance tools not preserved as specified for level A.
133.	Components not consolidated as specified for level A.
134.	Preservation and packing not in accordance with the referenced document for commercial.
135.	Engine and engine accessories not preserved as specified for commercial.
136.	Packing not as specified for level A or B.
137.	Depreservation guide not prepared and attached as specified for level A, B, or commercial.
138.	Marking illegible, incomplete, incorrect, or missing for level A, B, or commercial.

Source: https://assist.dla.mil -- Downloaded: 202610-18T03:03Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination and testing within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The first article pack may be prepared utilizing either the first article pump unit or a production pump unit. If the first article pump unit is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the first article pump unit and production pump units.

5.2 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly and matchmarking.

5.2.1.1.1 Disassembly. Disassembly of the pump unit shall be the minimum necessary to safeguard parts vulnerable to damage. Protruding parts that increase cubage which can be removed and reassembled without special tools shall be removed. All bolts and nuts, screws, pins, and washers removed shall be placed in mating parts secured to prevent their loss.

5.2.1.1.2 Matchmarking. Removed parts and mating parts shall be matchmarked to facilitate reassembly. Matchmarking identification shall be on shipping tags conforming to UU-T-81, type A, and the tags shall be attached to mating parts. Information on the tags shall be applied with waterproof ink.

5.2.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.3 Unprotected surfaces. Unpainted exterior metal surfaces of components of the pump unit requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.2.1.4 Engine. The engine and engine accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, type II, method I, except the engine cooling system shall be preserved by the water and antifreeze procedure using antifreeze conforming to MIL-A-46153.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

5.2.1.5 Pump. The pump shall be drained and the drain plugs coated with type P-10, grade 30 preservative before reinstalling. The suction and discharge ports shall be closed in a manner that will insure an oiltight seal. The air-vent valves shall be removed from the top of the pump head and not less than 1 gallon of type P-10, grade 30 preservative shall be poured through each opening. The valves shall be reinstalled. The oil plugs over the shaft seals shall be removed and type P-10, grade 30 preservative shall be applied to the carbon inserts. The oil plugs shall be coated with type P-10, grade 30 preservative and reinstalled. The pump shall be rotated with the engine starter for approximately 10 seconds and, after approximately 1 minute of standby, shall again be rotated to insure complete coverage of all interior surfaces of the pump. The preservative shall be left in the pump for shipment and storage. A tag conforming to UU-T-81, type A shall be attached indicating the pump is filled with preservative oil; drain and refill with the proper lubricant prior to operation. Information on the tag shall be applied with waterproof ink.

5.2.1.5.1 Pump openings. The suction and discharge openings shall be sealed with caps or plugs, as applicable, or with barrier material conforming to MIL-B-121, type II, grade A, class 1, secured in place with tape conforming to MIL-T-22085, type II.

5.2.1.6 Repair parts. Repair parts for each pump unit shall be preserved in accordance with the level A requirements of MIL-R-196. Parts not specifically covered in MIL-R-196 shall be preserved in accordance with the applicability, selection and application provisions of MIL-P-116.

5.2.1.7 Maintenance tools. Maintenance tools shall be preserved in accordance with PPP-P-40, level A.

5.2.1.8 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.9 Consolidation. Disassembled components, accessories, repair parts, maintenance tools, and technical publications for each pump unit shall be consolidated in the tool box as space permits. Items not fitting in the tool box shall be consolidated in a close-fitting box conforming to PPP-B-601, domestic type, style optional. The contents shall be secured within the box in a manner to prevent free movement and damage. Strapping shall not be required.

5.2.2 Commercial. Each complete pump unit with repair parts, tools, and publications, except for the engine and engine accessories (see 5.2.2.1), shall be preserved in accordance with ASTM D 3951.

Source: https://assist.dla.mil -- Downloaded: 20260-18T03:03Z
Check the source to verify that this is the current version before use.

5.2.2.1 Engine. The liquid cooling system shall be filled with a fresh clean solution of 50 percent water and 50 percent antifreeze conforming to MIL-A-46153. A tag shall be attached indicating the temperature to which the cooling system can be subjected to before damage. All lubrication reservoirs shall be filled with the lubricant or fluid designated for use. A tag or tags as applicable shall be attached in a conspicuous location indicating which lubricant or fluid has been used.

5.3 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

5.3.1 Level A. The pump unit, components, accessories, repair parts, maintenance tools, and technical publications comprising a complete unit shall be packed in a crate conforming to MIL-C-104, type I, class 1 or 2, style a. Blocking, bracing, and anchoring shall be in accordance with the appendix to the crate specification and ASTM D 3951.

5.3.2 Level B. Packing shall be as specified for level A, except that crates shall conform to MIL-C-3774, type I, style A. The contents of the crate shall be covered with a full shroud constructed and applied in accordance with the appendix to the crate specification.

5.3.3 Commercial. Pumps shall be packed in accordance with ASTM D 3951.

5.4 Depreservation guide. A depreservation guide shall be prepared, placed in a waterproof envelope marked "Depreservation Guide", and secured in a conspicuous location on the pump unit. Unless otherwise specified (see 6.2), DA Form 2258, "Depreservation Guide for Vehicles and Equipment", shall be used (see 6.9).

5.5 Marking.

5.5.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.5.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951. Additionally, each shipping container shall be marked with cube and gross weight.

6. NOTES

6.1 Intended use. The pump unit is intended for use either as a transfer pump or as a flood pump for use in petroleum products service.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval, and number of units required (see 3.2).

c. Color required (see 3.23).

d. When stenciling is to be other than specified (see 3.24).

e. When the welding process will not be at the option of the contractor (see 3.25.3.2).

f. Time frame required for submission of preproduction pack (see 5.1).

g. Degree of preservation and degree of packing required (see 5.2 and 5.3).

h. When other than DA Form 2258 is to be used (see 5.4).

6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction model pump unit. The preproduction model should consist of one or more units. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examinations, test, and approval of the first article.

6.4 Preproduction pack. Any changes or deviations of production packs from the approval preproduction pack will be subjected to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the pump units in accordance with this specification.

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each pump unit.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools, as necessary (including any special tools), and instructions regarding shipment of pump units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

6.7 Noise limits. When the noise level of the pump unit exceeds 85 dB(A), the appropriate discussion of the noise hazard in accordance with MIL-STD-1474 (MANUALS Paragraph) shall include the requirement for hearing protection, the noise level of the pump unit, the distance from the pump unit at which 85 dB(A) will always be met, and what operational situation or system configuration will reduce the level to 85 dB(A) (see 3.7.1).

6.8 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

6.9 DA Form 2258. When requested by the contractor, the contracting officer should arrange to furnish DA Form 2258.

Custodians:
Army - ME
Air Force - 99

Review activity:
DLA - CS
Air Force - 84

Preparing activity:
Army - ME

Project 4320-0293

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

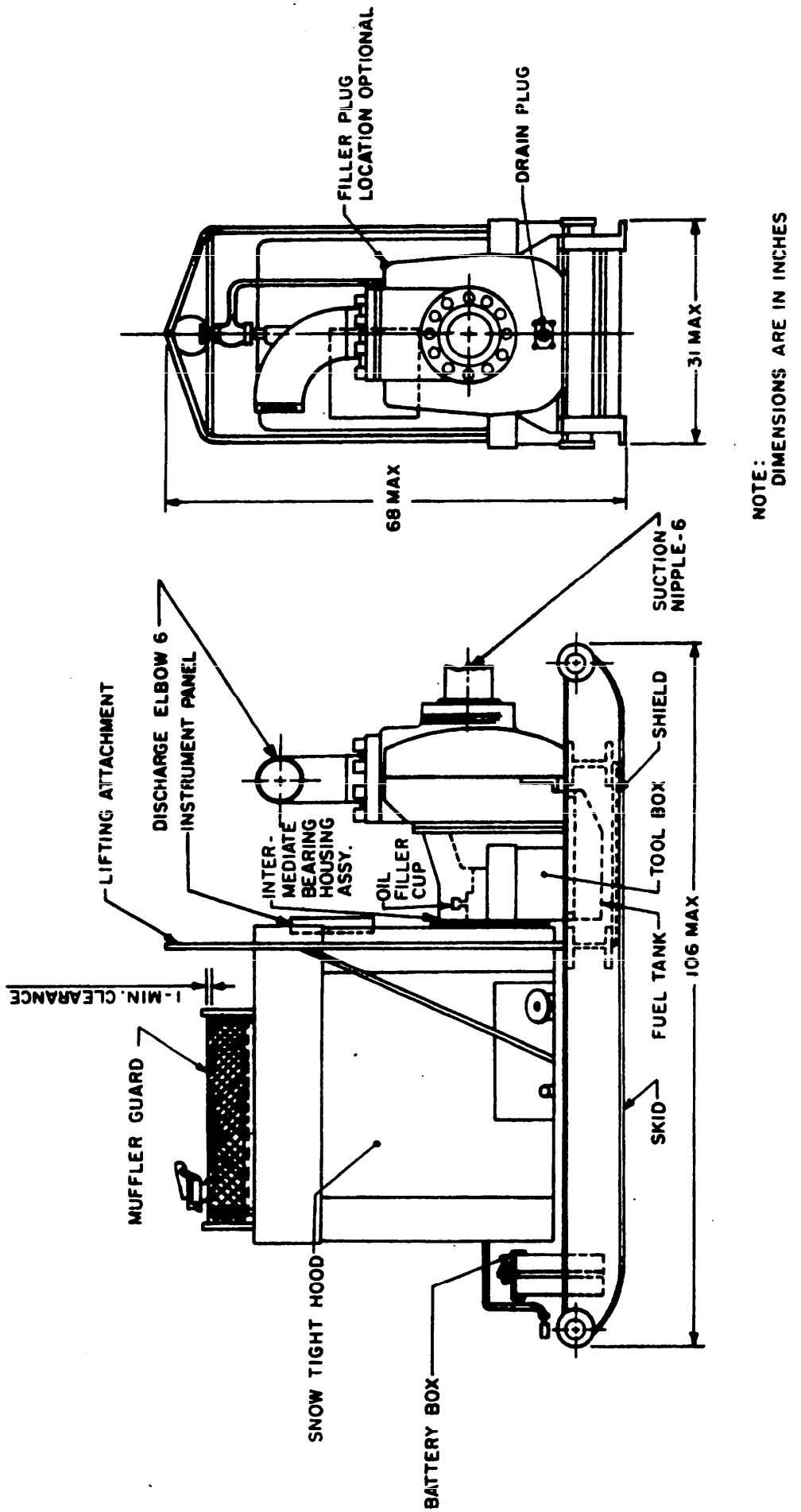
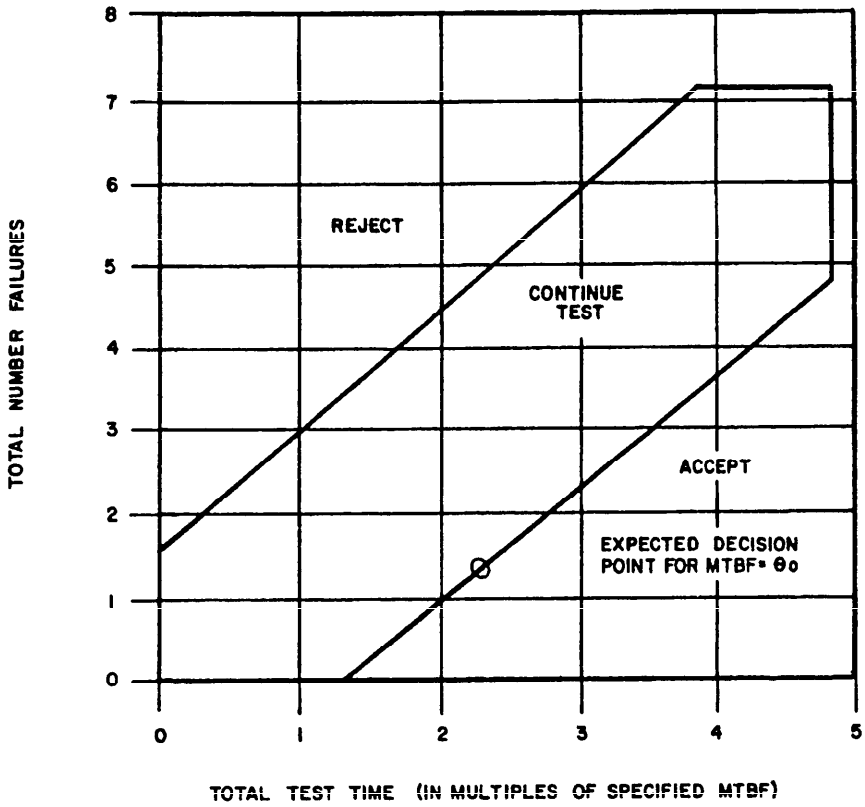


FIGURE 1. Pump, centrifugal: petroleum, diesel-engine driven, self priming, skid-mounted, single stage, 6-inch, grooved adapters, 245 gpm at 205-feet head to 1120 gpm at 100-feet head.

X-1328D

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

DECISION RISKS 20%
DISCRIMINATION RATIO 2.0:1



NO. OF FAILURES	TOTAL TEST TIME * REJECT (EQUAL OR LESS)	ACCEPT EQUAL OR MORE
0	N/A	1.40
1	N/A	2.09
2	.35	2.79
3	1.04	3.48
4	1.73	4.17
5	2.43	4.87
6	3.12	4.87
7	3.81	4.87
8	4.87	N/A

*TOTAL TEST TIME IS TOTAL UNIT HOURS OF "EQUIPMENT ON" TIME AND IS EXPRESSED IN MULTIPLES OF THE SPECIFIED MTBF.

FIGURE 2. Accept-reject criteria.

X-3661A

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Belvoir R&D Center
ATTN: STRBE-DS
Ft. Belvoir, VA 22060

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:03Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-14321E	Pump, Centrifugal: f/Petroleum Products, DED, Self Priming, Skid-Mtd, Single Stage; 6-In. Grooved Adapters, 245 GPM e

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* - Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* - Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* - Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-18T03:03Z
Check the source to verify that this is the current version before use.